

Faculty of Information Technology**Semester 1, 2026****FIT5145: Foundations of Data Science****Assignment 2: Description****Due Date: Friday, Week 8 (May 1, 2026), 11:55 PM**

The aim of this assignment is to investigate and visualise data using **R**. It will test your ability to:

1. Read data files and extract related data from those files;
2. Clean and process the data to ensure the results are valid.;
3. Perform exploratory data analysis and visualisation;
4. Use basic tools for managing and processing data; and
5. Communicate your findings in your report.

Assessment Details:

- Assessment Type: Individual Assignment
- Total marks: 15%
- Due Date: Friday, Week 8 (May 1, 2026), 11:55 PM. Please notice that we do not accept submissions after May 8, 2026 (i.e., 7 days after the due date).

Submission Details:

You are required to submit two separate files: a **PDF report** and an **RMD file**.

1. **PDF report:** Please convert (knit) your RMD file to HTML and then print the HTML file as a PDF. Your report must include *(a) code, (b) code output, and (c) an explanation* for each question.

(a) Code: For better readability, please use separate code chunks for each question or task.

(b) Code output: Please ensure that the outputs from your code and any written answers are clearly included for each question.

(c) Explanation: Please explain how you answered each question (i.e., explaining your codes or summarising your work for each question).
2. **RMD file:** This is the original R Markdown (RMD) file that you used to generate the PDF report. Please include both your code and explanations in this file.

Please do NOT zip your submission files. Zip file submissions will incur a 20% penalty on the assignment's total mark. Failure to submit any required files will result in a loss of marks for the evaluation of those files. **If you submit an incorrect version of a required file, you may request a resubmission; however, a late submission penalty will apply to the entire assignment to ensure marking fairness and consistency.** Please ensure that you submit the correct versions of all required files!

Notes:

1. Whenever you identify anomalies, errors, or any issues in the dataset that may negatively affect the validity of your analysis, **you are expected to determine and justify an appropriate approach, and perform the necessary data wrangling.**

This assignment also assesses your ability to critically evaluate data quality and make informed preprocessing decisions. Different approaches may be acceptable, provided they are clearly justified and appropriate for the analysis.

2. Whenever a question asks for a certain value, your code should produce the value. For example, when a question asks for the number of rows contained in a table, your code should print out the number of rows. **Extraction of the answer manually by eye-examination will not earn any marks.**
3. **Please check whether you can select and highlight texts in your PDF,** as shown below. This ensures the Turnitin score can be generated properly for your PDF file (we only need the Turnitin score for the PDF, not the RMD file).



Faculty of Information Technology

Semester 1, 2026

FIT5145: Foundations of Data Science

Assignment 2: Description

Due Date: Friday, Week 8 (May 1, 2026), 11:55 PM

The aim of this assignment is to investigate and visualise data using **R**. It will test your ability to:

1. Read data files and extract related data from those files;
2. Clean and process the data to ensure the results are valid;
3. Perform exploratory data analysis and visualisation;
4. Use basic tools for managing and processing data; and
5. Communicate your findings in your report.

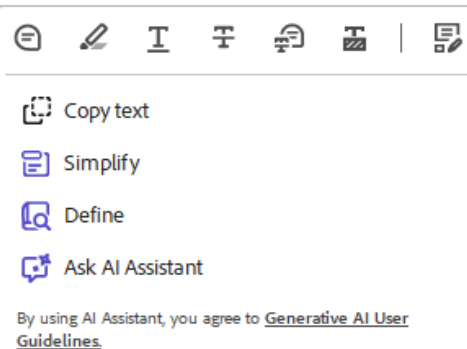
Assessment Details:

- Assessment Type: Individual A
- Total marks: 15%
- Due Date: Friday, Week 8 (May 1, 2026)
submissions after May 8, 2026

Submission Details:

You are required to submit two separate

1. **PDF report:** Please convert (using your favourite PDF tool) your R Markdown file to HTML and then print the HTML file as a PDF.



that we do not accept

Assignment Task:

The dataset for this assignment is in the file “ireland_news.csv”, and it can be accessed from the Assessments page on Moodle.

The data used in this task contains the news articles published by various news providers in Ireland from 1996 to 2021. It provides a long-term bird's-eye view of events in Europe. Therefore, analysing this dataset of news text is important for extracting insights such as trends, key events, and other interesting points related to news, news topics, and news providers.

Each row in the data file details information about a single news article, including the date of publication, its headline category (topic), the article's headline text, the corresponding news provider, and engagement score. The engagement score represents the level of user interaction with the article (e.g., clicks, views, shares), normalised to a value between 0 and 1, where higher values indicate greater engagement.

For instance, a sample row in the data is shown below, detailing the article's category as “news”, the publication date on 14/11/2009 by the Irish Times, the headline text: “Power supply lost to Dublin airport”, and an engagement score of 0.669.

publish_date	headline_category	headline_text	news_provider	engagement_score
Saturday, 14th of November, 2009	news	Power supply lost to Dublin airport	Irish Times	0.669
...	...	...	...	...

In this assignment, you are supposed to perform data analysis to gain a better understanding of the published news articles of different headline categories and news providers with different engagement scores in Ireland over different periods of time in the past years.

1. How many unique values are there in the headline_category column of the data file? (6 marks)
2. Which provider has the highest mean engagement? Which headline category has the lowest number of articles? (4 marks)
3. Compute the total number of articles for each headline category and news provider. Then, use a single R function to display the statistical information, i.e., Min, Max, and Mean, of the total number of articles (as computed previously) for each news provider. (Note: You may use multiple functions/commands to prepare the pre-processed data table, but when you compute and display the statistical information, you need to use a single R function.) (3 marks)
4. In which year did *TheJournal.ie* record its maximum engagement score? Which news provider shows the largest increase in engagement score over the years? (5 marks).
5. Investigate the factors associated with higher or lower engagement scores in the given dataset. Marks will be awarded based on the depth of your analysis, including discussion, and the strength of your reasoning (i.e., how well your insights and conclusions are supported). The maximum length is 5 pages; however, writing up to this limit does not necessarily result in higher marks. (5 marks)
6. Let's investigate headline categories, engagement scores, and time.
 - A. For each news provider, identify the top 3 headline categories with the strongest association between yearly article volume and yearly mean engagement score, based on correlation analysis. Please provide supporting results. (2 marks)
 - B. For each news provider, identify the top 3 headline categories with the most significant increasing trends in yearly articles counts. Please provide supporting results. (2 marks)
7. Please select the data for 2019 and 2020 and categorise it based on the period in which each record falls. To do this, create a new column named “**Period**” and assign appropriate period labels based on the **publish_date** value, as shown below:

publish_date	Period
1/1/19 - 31/3/19	Period 1
1/4/19 - 30/6/19	Period 2
1/7/19 - 30/9/19	Period 3
1/10/19 - 31/12/19	Period 4
1/1/20 - 31/3/20	Period 5
1/4/20- 30/6/20	Period 6
1/7/20 - 30/9/20	Period 7
1/10/20 - 31/12/20	Period 8

After that, compute the total number of articles by period and headline category, and generate a boxplot showing the distribution of the total number of articles for each period. (3 marks)

8. We want to examine the trend in the number of articles published by RTE News in September over the years. Please create an appropriate chart. (4 marks)
9. Using both the original dataset and external datasets, investigate the factors influencing the yearly trend in the number of articles published by some news providers. You may select one or more news providers for this investigation and should analyse at least 20 years of data. Marks will be awarded based on the depth of your analysis (including discussion) and the strength of your reasoning (i.e., how well insights and conclusions are supported). **Note:** Please provide source links for any external data used. The maximum length is 6 pages; however, writing up to this limit does not necessarily result in higher marks. (7 marks)